Lecture 7

The Fourier transform

Objectives. *On the whole space, Fourier coefficients become a function of a continuous wave number. We prove inversion by inserting a Gaussian factor and then removing it. Inversion gives an isometry on rapidly decreasing smooth functions. Density extends this map to a unitary map of $L^2(\mathbb{R}^d)$. The same Gaussian argument proves uniqueness for integrable functions.*

By the end of this lecture you will be able to:

- Check Fourier signs and constants for translation, modulation and differentiation.
- Prove that the Fourier transform preserves the Schwartz space.
- Derive inversion and the unitary extension onto $L^2(\mathbb{R}^d)$.
- Prove convolution and uniqueness for L^1 functions.

7.1 The transform of an integrable function

Fix $d \in \mathbb{N}$. Position is $x \in \mathbb{R}^d$, and wave number is $k \in \mathbb{R}^d$. For $f \in L^1(\mathbb{R}^d)$ define

$$\widehat{f}(k) = (\mathcal{F}f)(k) = \frac{1}{(2\pi)^{d/2}} \int_{\mathbb{R}^d} e^{-\mathrm{i}k\cdot x} f(x)\,\mathrm{d}x. \tag{7.1}$$

Absolute integrability makes this well-defined at every k and gives $\left\|\widehat{f}\right\|_\infty \le (2\pi)^{-d/2}\,\|f\|_1$.

Proposition 7.1. *For $f \in L^1(\mathbb{R}^d)$, its transform is uniformly continuous and tends to zero as $|k| \to \infty$.*

Proof. The difference at $k+h$ and k is bounded, uniformly in k, by

$$\frac{1}{(2\pi)^{d/2}} \int_{\mathbb{R}^d} |e^{-\mathrm{i}h\cdot x} - 1||f(x)|\,\mathrm{d}x.$$

Dominated convergence makes this tend to zero as $h \to 0$. For the decay, set $\tau_a f(x) = f(x-a)$. Change of variable gives $\widehat{\tau_a f}(k) = e^{-\mathrm{i}k\cdot a}\widehat{f}(k)$. When $k \neq 0$, choose $a = \pi k/|k|^2$. Then

$$2|\widehat{f}(k)| \le (2\pi)^{-d/2}\,\|f - \tau_a f\|_1 \longrightarrow 0,$$

since $a \to 0$ as $|k| \to \infty$ and translations are continuous in L^1 by Theorem 5.5. ∎

This decay statement is the *Riemann–Lebesgue lemma*. The same substitution gives the translation and modulation rules

$$f(x-a) \longmapsto e^{-\mathrm{i}k\cdot a}\widehat{f}(k), \qquad e^{\mathrm{i}b\cdot x} f(x) \longmapsto \widehat{f}(k-b), \qquad a, b \in \mathbb{R}^d.$$

For modulation, multiply the two exponential factors to obtain $e^{-\mathrm{i}(k-b)\cdot x}$ under the defining integral. If also $x_j f \in L^1$, differentiation under that integral gives

$$\widehat{x_j f} = \mathrm{i}\partial_{k_j}\widehat{f}. \tag{7.2}$$

The bound $|x_j f|$ justifies the derivative by the parameter-integral theorem, Theorem 4.4. For example, translating a Gaussian changes its transform only by a factor of absolute value one. Modulating it translates its transform in wave number.

7.2 Schwartz space and differentiation

Recall the multi-index notation: for $\alpha \in \mathbb{N}_0^d$, $|\alpha| = \sum_j \alpha_j$, $x^\alpha = \prod_j x_j^{\alpha_j}$ and $\partial^\alpha = \prod_j \partial_{x_j}^{\alpha_j}$.

Definition 7.2. The *Schwartz space* $\mathcal{S}(\mathbb{R}^d)$ consists of smooth functions f whose seminorms

$$\|f\|_{\alpha,\beta} = \sup_{x\in\mathbb{R}^d} |x^\alpha \partial^\beta f(x)|$$

are finite for all $\alpha, \beta \in \mathbb{N}_0^d$. Convergence in this space means convergence in every such seminorm.

This is the space used for density in Theorem 5.8. Multiplying a derivative by $(1+|x|^2)^M$ and expanding that polynomial shows that each derivative decays at least as $(1+|x|^2)^{-M}$, for every integer $M \geq 0$. Taking $2M > d$ proves that every derivative, and every polynomial times a derivative, is in L^1 and L^2.

For example, differentiating $e^{-|x|^2/2}$ repeatedly gives a polynomial times the same Gaussian. It belongs to $\mathcal{S}$. By Fubini (Theorem 4.1) and Theorem 4.5, its transform is

$$\mathcal{F}(e^{-|x|^2/2})(k) = e^{-|k|^2/2}. \tag{7.3}$$

The one-dimensional factors are those of Lecture 4, each with the normalising factor $(2\pi)^{-1/2}$.

Theorem 7.3. *The transform maps $\mathcal{S}(\mathbb{R}^d)$ linearly and continuously into itself. For $f \in \mathcal{S}$,*

$$\widehat{\partial_j f} = \mathrm{i}k_j \widehat{f}, \qquad \widehat{x_j f} = \mathrm{i}\partial_{k_j}\widehat{f}.$$

Here continuity means that each output seminorm is bounded by a constant times a finite sum of input seminorms.

Proof. Every moment is integrable, so repeated differentiation under the integral makes $\widehat{f}$ smooth. Integration by parts in x_j gives the first rule: the boundary term is zero by rapid decay, and differentiating the kernel gives $-\mathrm{i}k_j e^{-\mathrm{i}k\cdot x}$. Fubini justifies integrating the remaining coordinates. The second rule is (7.2). Repeating the two rules gives

$$k^\alpha \partial_k^\beta \widehat{f} = (-\mathrm{i})^{|\alpha|+|\beta|}\mathcal{F}\big(\partial_x^\alpha (x^\beta f)\big).$$

Its supremum is bounded by $(2\pi)^{-d/2}$ times the L^1 norm of the function inside the transform. The product rule expresses that function as a finite sum of polynomial multiples of derivatives of f. For each such term g, choose $2M > d$ and use

$$\|g\|_1 \leq \sup_x |(1+|x|^2)^M g(x)| \int_{\mathbb{R}^d} (1+|x|^2)^{-M}\,\mathrm{d}x.$$

Expanding the polynomial bounds the supremum by finitely many input seminorms. The integral is finite, for example by the radial formula in Theorem 4.2. This proves the stated estimate and continuity. ∎

Jacob–Evans IV, Theorem 11.3, pp. 173–174, uses exactly this transform normalisation and proves the same seminorm estimate.[1]

7.3 Inversion by Gaussian regularisation

For $g \in \mathcal{S}$ define the inverse-sign integral by

$$(\mathcal{F}^{-1}g)(x) = \frac{1}{(2\pi)^{d/2}} \int_{\mathbb{R}^d} e^{\mathrm{i}k\cdot x} g(k)\,\mathrm{d}k.$$

At this point the notation denotes an integral, whose inverse property we will prove. It equals $\widehat{g}(-x)$, so it preserves $\mathcal{S}$ and satisfies the same continuity estimates.

For $\varepsilon > 0$ the normalised Gaussian kernel is

$$H_\varepsilon(y) = \frac{e^{-|y|^2/(2\varepsilon)}}{(2\pi\varepsilon)^{d/2}}, \qquad \int_{\mathbb{R}^d} H_\varepsilon = 1.$$

The substitution $y = \sqrt{\varepsilon}\,z$ proves both the mass identity and $\int_{|y|\ge\delta} H_\varepsilon \to 0$ for each $\delta > 0$. Thus these kernels are an approximate identity.

Lemma 7.4. *For $f \in L^1(\mathbb{R}^d)$, $\varepsilon > 0$ and $x \in \mathbb{R}^d$,*

$$\frac{1}{(2\pi)^{d/2}} \int_{\mathbb{R}^d} e^{\mathrm{i}k\cdot x} e^{-\varepsilon|k|^2/2} \widehat{f}(k)\,\mathrm{d}k = (f * H_\varepsilon)(x). \tag{7.4}$$

Proof. Insert the integral defining $\widehat{f}$. Fubini (Theorem 4.1) applies because the absolute double integral is bounded by $\|f\|_1 \int e^{-\varepsilon|k|^2/2}\,\mathrm{d}k < \infty$. In each coordinate, substitute $k_j = \kappa_j/\sqrt{\varepsilon}$, with $\kappa_j \in \mathbb{R}$, and apply Theorem 4.5 at wave number $-(x_j - y_j)/\sqrt{\varepsilon}$. This gives the inner integral

$$\frac{1}{(2\pi)^{d}} \int_{\mathbb{R}^d} e^{\mathrm{i}k\cdot(x-y)} e^{-\varepsilon|k|^2/2}\,\mathrm{d}k = H_\varepsilon(x-y).$$

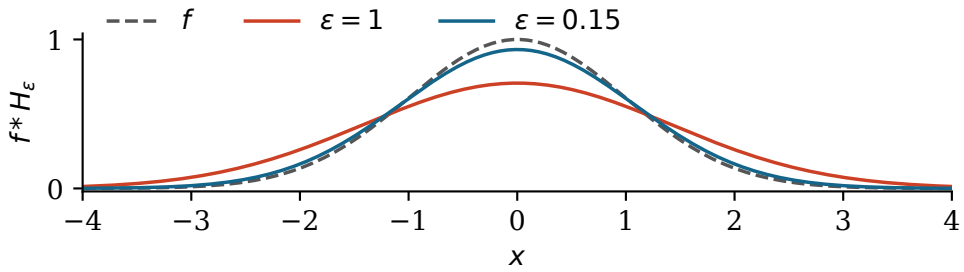
Integration against $f(y)$ gives the claimed convolution. It exists at every x because H_ε is bounded. ∎

For the Gaussian $f(x) = e^{-x^2/2}$, completing the square gives

$$(f * H_\varepsilon)(x) = \frac{1}{\sqrt{1+\varepsilon}} e^{-x^2/(2(1+\varepsilon))}.$$

This example illustrates removal of the regularisation factor.

[1] Their transform variable ξ is written k here. No constant or sign conversion is needed.



Gaussian convolutions for $\varepsilon = 1$ and 0.15, with the limiting function $e^{-x^2/2}$ dashed. The curves use the exact convolution formula.

Theorem 7.5 (Fourier inversion). *The transform is a continuous bijection of $\mathcal{S}(\mathbb{R}^d)$ with continuous inverse $\mathcal{F}^{-1}$. For every $f \in \mathcal{S}$ and $x \in \mathbb{R}^d$,*

$$f(x) = \frac{1}{(2\pi)^{d/2}} \int_{\mathbb{R}^d} e^{\mathrm{i}k\cdot x} \widehat{f}(k)\, \mathrm{d}k. \tag{7.5}$$

Moreover, $\mathcal{F}^2 f(x) = f(-x)$ and $\mathcal{F}^4 f = f$.

Proof. The transform $\widehat{f}$ is Schwartz and hence integrable. Dominated convergence takes the left side of (7.4) to $\mathcal{F}^{-1}\mathcal{F} f(x)$. The right side tends uniformly to f: split $\int H_\varepsilon(y)[f(x-y) - f(x)]\,\mathrm{d}y$ at $|y| = \delta$. Uniform continuity makes the inside part arbitrarily small, and the outside part is bounded by $2\|f\|_\infty \int_{|y|\ge\delta} H_\varepsilon$. Hence $\mathcal{F}^{-1}\mathcal{F} f = f$.

In this proof, let R denote reflection, given by $Rf(x) = f(-x)$. Substitution in the transform gives $\mathcal{F}R = R\mathcal{F}$, and the inverse-sign integral is $R\mathcal{F}$. Thus the identity just proved is $R\mathcal{F}^2 = 1$, so $\mathcal{F}^2 = R$ and $\mathcal{F}^4 = 1$. It also gives $\mathcal{F}\mathcal{F}^{-1} = \mathcal{F}R\mathcal{F} = R\mathcal{F}^2 = 1$. Both maps already satisfy the seminorm continuity estimates. ∎

Corollary 7.6. *For $f \in \mathcal{S}(\mathbb{R}^d)$, $\int_{\mathbb{R}^d} \widehat{f}(k)\,\mathrm{d}k = (2\pi)^{d/2} f(0)$.*

This is (7.5) evaluated at zero.

7.4 Plancherel and the unitary extension

A linear map between Hilbert spaces is *unitary* if it preserves inner products and is onto. Onto is a separate requirement from norm preservation.

Theorem 7.7 (Plancherel). *The transform has a unique extension to a unitary map $\mathcal{F} : L^2(\mathbb{R}^d) \to L^2(\mathbb{R}^d)$. For every $f, g \in L^2$,*

$$\left\langle \widehat{f}, \widehat{g} \right\rangle = \langle f, g \rangle, \qquad \left\| \widehat{f} \right\|_2 = \|f\|_2 .$$

On $L^1 \cap L^2$, the extension agrees almost everywhere with the integral transform. On general L^2 functions, the defining integral need not exist.

Proof. First take $f, g \in \mathcal{S}$. Expanding $\overline{\widehat{f}}$, Fubini and inversion give

$$\begin{aligned}\left\langle \widehat{f}, \widehat{g} \right\rangle &= \int_{\mathbb{R}^d} \overline{f(x)} (\mathcal{F}^{-1}\widehat{g})(x)\, \mathrm{d}x \\ &= \langle f, g \rangle .\end{aligned}$$

Fubini is justified by $\|f\|_1 \, \|\widehat{g}\|_1 < \infty$. In particular the transform preserves the L^2 norm on $\mathcal{S}$.

Next let $f \in L^2$. Choose $f_m \in \mathcal{S}$ with $f_m \to f$ in L^2, using Theorem 5.8. The isometry makes $(\widehat{f_m})$ Cauchy in L^2. Completeness supplies a limit, which defines $\mathcal{F} f$. If another sequence approximates f, the norm of the difference of the transforms equals that of the approximants and tends to zero. The extension is therefore well-defined and unique. Passing to limits in inner products proves the asserted identity for all $f, g \in L^2$.

Its range is closed: a convergent sequence $\mathcal{F} f_m$ makes f_m Cauchy by the norm identity, and its limit maps to the given limit. Inversion on $\mathcal{S}$ shows that the range contains $\mathcal{S}$. This subspace is dense in L^2, so the closed range is all of L^2.

Finally suppose $f \in L^1 \cap L^2$. Truncate it to $f \mathbb{1}_{\{|x| \le m,\, |f| \le m\}}$ and convolve the truncation with a compactly supported smooth mollifier from Lecture 5. Choose the mollifier scale so the error is below $1/m$ in both L^1 and L^2, using Theorem 5.7 for each norm. The resulting functions g_m are compactly supported and smooth: bounded derivatives of the mollifier may be passed under the integral against the compactly supported L^1 truncation. They tend to f in both norms, because truncation does so by dominated convergence. Their integral transforms converge uniformly to the integral transform of f, by the L^1 bound. Their transforms also converge in L^2 to the extension. An almost-everywhere convergent subsequence of the latter convergence, supplied by Theorem 5.3, identifies the two limits. ∎

For example, the self-dual Gaussian has the same norm in position and wave number. More generally, Plancherel defines wave-number amplitudes even for square-integrable functions that are not integrable. QM-II proves the position–momentum uncertainty relation on Schwartz functions in its uncertainty theorem. This result is not used here.

7.5 Convolution and uniqueness

Theorem 7.8. *For $f, g \in L^1(\mathbb{R}^d)$, the convolution is integrable and*

$$\widehat{f * g} = (2\pi)^{d/2} \widehat{f}\, \widehat{g}.$$

For $f, g \in \mathcal{S}$, there is also the product formula $\widehat{fg} = (2\pi)^{-d/2} \widehat{f} * \widehat{g}$.

Proof. Tonelli gives $\|f * g\|_1 \le \|f\|_1 \, \|g\|_1$. In its transform substitute $x = y + z$ and use Fubini:

$$\begin{aligned}\widehat{f * g}(k) &= \frac{1}{(2\pi)^{d/2}} \left(\int_{\mathbb{R}^d} e^{-\mathrm{i}k \cdot y} f(y)\, \mathrm{d}y \right) \\ &\quad \times \left(\int_{\mathbb{R}^d} e^{-\mathrm{i}k \cdot z} g(z)\, \mathrm{d}z \right) \\ &= (2\pi)^{d/2} \widehat{f}(k) \widehat{g}(k).\end{aligned}$$

For the product formula substitute the inverse integral for g into the transform of fg. Fubini applies by $\|f\|_1 \, \|\widehat{g}\|_1 < \infty$. The inner integral is $(2\pi)^{d/2}\widehat{f}(k-q)$, leaving $(2\pi)^{-d/2}\int \widehat{f}(k-q)\widehat{g}(q)\,\mathrm{d}q$. ∎

Theorem 7.9 (Uniqueness in L^1). *If $f \in L^1(\mathbb{R}^d)$ and $\widehat{f} = 0$, then $f = 0$ almost everywhere.*

Proof. The regularisation identity (7.4) holds for every L^1 function, independently of inversion. If $\widehat{f} = 0$, it gives $f * H_\varepsilon = 0$ at every point. By Theorem 5.7, these convolutions tend to f in L^1. Hence $\|f\|_1 = 0$, as required. ∎

In particular, two integrable functions with the same transform agree almost everywhere, by applying uniqueness to their difference.

The unitary extension discharges QM-II's standing fact S1. Poisson summation is assigned to Sheet 7. Its proof will be given in Lecture 22.

Summary.

- Gaussian regularisation proves inversion on $\mathcal{S}$ and uniqueness on L^1.
- Inversion makes the L^2 extension onto, as well as an isometry.
- For two L^1 functions, unnormalised convolution contributes the factor $(2\pi)^{d/2}$ to the transform of their convolution.
- The transform preserves Schwartz space continuously in both directions.

References

N. Jacob and K. P. Evans, *A Course in Analysis IV*, World Scientific, 2018, Chapter 11, pp. 173–179. Its equation (11.1) uses the same unitary normalisation as (7.1).